

HR experts nowadays focus on optimizing the mixture of human and automatic paintings to advantage simple, seamless, and intuitive paintings surroundings. It affords them time for creativity, intelligence, and empathy to supply more desirable candidates and worker experience (Babasaheb, Sphurti & Khang, 2023).

Consequently, the term “synthetic intelligence” is carried out while a device mimics “cognitive” features of human beings partnered with different human minds, such as “getting to know” and “problem-solving.”

Modern system competencies normally labeled as AI encompasses effectively expertise human speech, competing to the best degree in strategic recreation systems (consisting of games such as chess and Go), autonomously working cars, and shrewd routing in content material shipping networks and army simulations.

According to Kaplan and Haenlein (2018), three extraordinary forms of AI systems are: analytical, human-stimulated, and humanized synthetic intelligence.

Analytical AI has the most effective traits, with cognitive intelligence producing a cognitive illustration of the sector and the usage of study primarily based totally on beyond revel in to tell destiny decisions. Human-stimulated AI has factors from cognition in addition to emotional intelligence and expertise, similarly to cognitive factors, additionally human feelings recall of their decision-making (Rana et al., 2021).

AI that has been “humanized” demonstrates characteristics of all intelligence types (cognitive, emotional, and social), and is able to be self-aware and self-conscious when interacting with people.

It’s simple to overlook the significant role that the same technologies play in locating, securing, and keeping personnel even while AI and automation may eventually replace certain roles. Finding the proper people is harder than ever in this era of rapid change and a lack of digital capabilities. Businesses can swiftly and simply find a wide range of top prospects utilizing AI and automation at a rate that keeps up with the frenetic pace of modern business (Khanh & Khang, 2021).

Chatbots and automated social media scraping tools are two examples of elementary AI systems that can aid recruiters in the sourcing and selection procedures. These instruments are intended to offer mediocre or subpar indicators of an applicant’s chances of success at the organization.

Tests and simulations that directly collect candidate data are forms of intermediate artificial intelligence applications that are utilized in the hiring process (Rani et al., 2021).

One corporation that has made use of an efficient AI screening procedure is Unilever, which was created with the assistance of its partner, Pymetrics. Candidates are required to play neuroscience-based Pymetrics games for about 20 minutes as part of the hiring strategy. These scientific games covertly measure a candidate’s memory, response to risk, comprehension of contextual cues, and attentiveness.

Although these predictors typically don’t concentrate on specific job metrics, they give recruiting managers a reasonably good sense of whether an applicant is a suitable fit for the role or not. However, cutting-edge AI systems make use of unique algorithms that were developed, especially to connect certain job performance metrics to possible applicants who best show these traits.

AI software developed by HireVue examines each applicant’s responses, body language, emotional condition, tone, and keywords. Additionally, Affectiva has recently created emotion-recognition software that aids in determining a candidate’s

sincerity and emotional quotient. Because these AI software tools are so sophisticated, they can really “catch the spark of hatred that passes across an applicant’s face when he talks his ex-boss,” according to a study (Nicastro, 2017; WirePiazza, 2018).

In HR management, AI’s effect is becoming more and more. It begins with the recruiting process and ends with the employee’s departure. It involves things like training, engagement, benefits, records, and so on. Here are a handful of the influences that are emphasized. This chapter discusses a few difficulties that arise when employing artificial intelligence for HR administrative duties (Lawler & Elliot, 1996; Basu et al., 2023).

3.2 AI AND HR FUNCTIONS

3.2.1 PERSONALIZED EMPLOYEE EXPERIENCES

In their study, IBM officials covered how AI may be successfully included into the onboarding process for new employees. New hires may not know where to go when they want to network and learn more about the company. They might speak with their desk neighbors. But what if that person is employed by a separate division and what if an employee had received welcoming materials for new hires that were specific to his first job on his mobile device? It is in the paper on changing HR with AI, IBM officials wrote (Snehal, Babasaheb & Khang, 2023).

In order to help new hires get up to pace swiftly, IBM is working to develop a system that will respond to their most important or job-critical queries. For instance, an AI may offer training recommendations or provide names, locations, and contact information (Palos-Sánchez et al., 2022).

3.2.2 COGNITIVE-SUPPORTING DECISION-MAKING

Officials from IBM, who are obviously marketing their own AI capabilities through IBM Watson, also showed off how cognitive engines could aid employees in making important day-to-day decisions at work (Babasaheb, Sphurti & Khang, 2023). Typically, the HR team members would be responsible for these duties:

- **Vacation requests** – Employees who wish to ask for time off are warned that their request is unlikely to be granted because so many other people have already reserved time off during that period.
- **Mood assessment** – A worker answers a client call. Following the call, the employee is advised to take a break before his meeting because he appears worried.
- **Team training** – Team managers are given a list of training possibilities for team members when an organization wants to take a more systematic approach to employee training.
- **Recruiting procedures** – A hiring manager is informed that the company’s recruitment strategy fails because too few prospects are interviewed. In addition to other things, cognitive solutions can help firms to gain access to various data sources, revealing fresh insights that can be used to construct applicant profiles.

3.2.3 AUTOMATING “REPETITIVE, LOW-VALUE ADD TASKS”

AI offers HR the chance to automate “repetitive, low-value add tasks” and put greater emphasis on more strategic work (Tailor, Pareek & Khang, 2022). Economy time in those areas can assist HR teams in refocusing on “value-add activities,” such as mentorship and ongoing feedback. HR spends time processing the stages of onboarding a new employee. AI solutions perform routine HR operations, such as benefits administration and query triage (Giraud et al., 2022).

3.2.4 ANALYTICS FOR SMARTER PEOPLE

Businesses have been gathering data about their clients for years in order to glean insights and forecast future behavior. In terms of utilizing these employee analytics, HR departments still have a long way to go (Khang, Chowdhury & Sharma, 2022).

AI will play a crucial role in HR after it is decided what data to track, examine, manage, and secure. Companies will search for creative ways to attract top people in the never-ending search for talent. Companies will be able to stand out from one another with the use of technologies that improve the candidate’s experience and satisfy the candidate’s digital expectations (Bhambri et al., 2022).

3.2.5 GETTING RID OF BIASES

HRPA poll revealed that, despite their best efforts to be inclusive, some employers may unconsciously favor applicants who are most like them, which it calls “unconscious bias.”

The Implicit Association Test (IAT), a psychological technique that demonstrates that people’s unconscious word associations reflect bias, has shown yet another bias, language bias. These prejudices creep into résumé evaluations and job descriptions. Thanks to AI, organizations can now use algorithms to discover and eliminate these bias patterns in their hiring messages and welcome diverse applications.

Additionally, AI could offer managers candidates who may have been passed over because of the propensity of humans to favor candidates with similar traits.

3.2.6 EMPLOYEES WHO MAY BE LEAVING

Veriato’s AI platforms are built to monitor employee computer activity, including emails, keystrokes, and web browsing, and retain them for a month. The platforms use an AI system to evaluate the information to establish a baseline for typical activity patterns inside the company.

According to HRPA researchers, “based on that knowledge, it flags outliers and rumor them to the employer and also detects changes in the whole nature of employees’ communications to anticipate when individuals might be considering leaving.”

3.2.7 LITIGATION APPROACH

The fact-driven nature of employment-related disputes makes it crucial to collect supporting documentation and other facts. There is only 5% of survey participants from littler are utilizing advanced analytics to direct their litigation strategy.

According to John et al. (1996), employers might not be aware of the advantages of adopting analytics in this situation. It is revolutionary to be able to use data early in a case to elicit ideas before taking a deposition or starting to evaluate the consistency of witnesses.

Crews added that finding important documents enables the employer to understand what individuals were actually doing at a particular moment and can aid in the creation of a narrative.

3.2.8 PAY EQUITY

Pay equity can also be evaluated using data analytics. State and local legislation in this area is evolving quickly. For instance, rules preventing employers from inquiring about job candidates' former salaries have been passed in at least 12 jurisdictions. Such laws aim to end the ongoing practice of historically discriminatory pay practices based on gender, race, and ethnicity. Technology can be very useful in detecting differences in employee salary based on protected groups.

HRPA noted that there are technologies that make it simple to create a user-friendly experience as well as to evaluate, comprehend, and transmit data. You are no longer limited to using an Excel file with lots of arithmetic. It is useful to gain an understanding of what is truly happening within the organization.

3.2.9 CHATBOTS

Using certain technology, like chatbots, employees can obtain crucial information on policies and processes whenever and wherever they want. Chatbots can be beneficial for providing typical employee inquiries with text-based responses.

According to a 2017 poll of 350 HR directors, conducted by the Santa Clara, CA-based startup ServiceNow, two-thirds of respondents said they think employees are more comfortable utilizing chatbots than other forms of interaction for transactional questions about paid-time-off policy, open enrollment, and leaves of absence.

ServiceNow specializes in cloud computing. Employers who utilize chatbots must make sure they are adhering to all applicable federal, state, and local employment regulations regarding data security (Khang et al., 2022b), disability, and other matters (Rani et al., 2023).

3.2.10 LEGAL PITFALLS

HR practitioners must check systems for bias when utilizing AI to guide HR strategy. They should watch out for disparate impact, which occurs when a standard appears to be fair or neutral in theory but is actually discriminatory in practice.

For instance, a recruiting tool might eliminate applicants who live more than 10 miles from the workplace. What if wealthy white families predominate in the neighborhoods close to the workplace? Based on race and ethnicity, this employment criterion could have an unequal effect (Priksht et al., 2023; Pereira et al., 2023).

3.2.11 RECRUITING

A lot of our decisions are intuitive. According to one survey, the majority of hiring managers decide on a candidate within the first 60 seconds of seeing them, frequently on the basis of appearance, handshake, dress, or speech.

Do we actually understand the qualities, backgrounds, degrees, and personality traits that make someone successful in a particular role? Not at all. Managers and recruiting professionals spend billions on assessments, tests, simulations, and games to find new employees, but many admit to making mistakes with 30–40% of their picks.

AI-based algorithms can sort through resumes, identify qualified employees, profile top performers, and even decode video interviews to provide us with signals about which individuals are most likely to succeed, as shown in Figure 3.1.

AI in hiring will be very important. While technical skills account for a relatively minor portion of a person’s success, we are all concerned about job skills (such as software, sales, and arithmetic).

The level 4 maturity companies, those with the highest financial return from hiring, assigned nearly 40% of their hiring criteria to emotional and psychological traits, such as ambition, learning agility, passion, and sense of purpose according to some of the most recent studies on high-impact talent acquisition.

3.2.12 MANAGEMENT AND CONTROL

We behave in a Zen-like manner. We read books, attend workshops, imitate our favorite bosses, and extol the virtues of the current crop of prosperous leaders. Are we really experts in the field of leadership? It’s a transient subject, in the authors’ opinion.

This year, we are focused on purpose, mission, and followers in a few years ago, it was “servant leadership” and when you were young it was “execution and financial acumen.” Most studies find that there are dozens of management and leadership

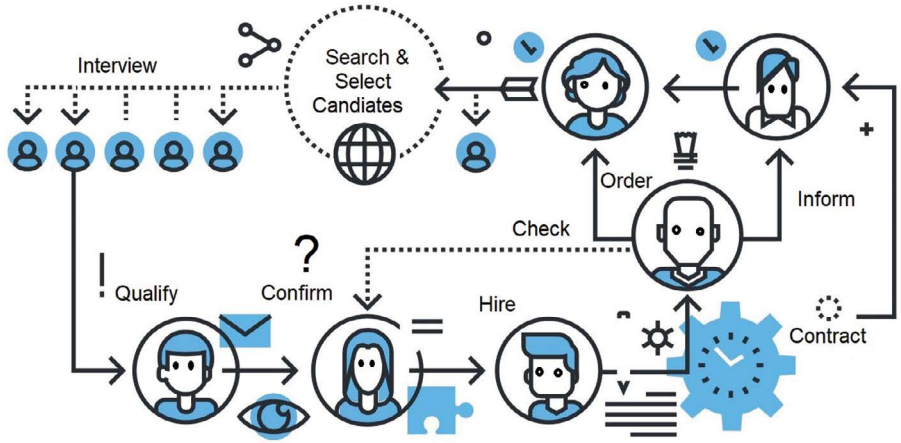


FIGURE 3.1 Illustration of recruitment workflow in the real world.

traits that define success, and each of us brings a slightly different and unique combination of them.

3.2.13 FRAUD AND COMPLIANCE

According to some recent studies, there are numerous employees who steal or commit crimes are “contagious” to their coworkers (i.e., people who work with them pick up bad habits). AI may analyze organizational network data (email traffic, sentiment of comments). It may detect stress points, potential ethical lapses, and many other compliance risks, and then flag the “red areas” for HR or compliance professionals to address before bad behavior takes place.

3.2.14 WELFARE AND EMPLOYEE COMMITMENT

AI is increasingly being used to pinpoint behaviors that lead to subpar performance at work. AI can recognize actions and circumstances that increase the risk of accidents. A new generation of survey tools can spot stress and unruly behavior trends and inform HR or line supervisors. Limeaid, VirginPulse, Glint, Ultimate Software, CultureAmp, TinyPulse, Peakon, and other vendors are among those in this market.

3.2.15 EMPLOYEE SELF-SERVICE AND CANDIDATE MANAGEMENT

Conversations can be clever and simple thanks to a new breed of intelligent chatbots. IBM, ServiceNow, Xor, Mya, Ideal, Paradox, and more vendors are operating in this market.

3.3 RESULTS AND DISCUSSION

AI consists of a variety of algorithms and machine learning technologies that can quickly process data, find patterns, optimize, and predict trends. It is not some mystical computerized avatar. The systems are capable of comprehending speech, recognizing images, and using pattern-matching to detect signals indicating emotion, sincerity, and even personality.

Although these algorithms lack the “intuition” of humans, they are fast, analyzing millions of pieces of data in a matter of seconds and instantly correlating them with patterns (Rani et al., 2022).

By drawing curves representing potential outcomes and then optimizing choices depending on a variety of factors, statistically speaking, AI systems may “predict” and “learn.” Consequently, one may envision an AI system that examines a candidate’s potential demographics, employment history, and interview questions beforehand.

These are all brand-new uses, and despite how fascinating they may seem, there are several dangers to consider. The greatest is that AI needs “training data” in order to function. Or put another way, algorithms pick up knowledge from the past.

If your existing management procedures are unfair, punitive, unduly hierarchical, or biased, you can end up institutionalizing everything you despise (Khang, Rani & Sivaraman, 2022).

We require “tuneable” AI that is transparent so that we can check the algorithms’ actions to make sure they are correct. Our early algorithms will require “bumpers” and “tuning knobs,” as early automobiles didn’t always drive straight, and we will need to learn how to improve them.

Bias may be institutionalized by the systems. Imagine that your company has a low percentage of African American engineers and has never employed any women in engineering. The AI hiring process would logically deduce that women and minority engineers had lower managerial aspirations. It will take time to effectively remove this kind of bias from the algorithms, which must be done cautiously.

Data exposure and unintentional exploitation are both possible risks. Think about a typical use of analytics where we attempt to foretell the possibility that a high performer will leave the firm. If we warn managers that a certain employee is likely to leave, we risk encouraging the wrong behavior—the manager might ignore the employee or treat him or her differently.

We must carefully study how to use behavioral economics. Today, AI is a “tool” for recommendation and improvement rather than a self-contained system of making decisions.

The necessity to develop “interpretive” and “transparent” AI systems was explored by leaders in the field at Entelo. To put it another way, any time the system decides something, it should explain why it did so, so that people may assess if the criteria it employed are still valid. This is one of the most crucial requirements for new technologies, but sadly, the majority of AI systems are now a total mystery.

The buzz surrounding AI is at an all-time high right now. Every vendor of HR software wants you to think they have a machine learning team and the best AI solution available.

Yes, these talents are crucial to this sector, but don’t buy into the marketing hype.

The effectiveness of an HR tool will depend on a number of factors, including the completeness and accuracy of its algorithms, the usability of its systems, but more importantly, it will depend on its capacity to offer “narrow AI,” or extremely precise solutions that address your concerns. Only when the vendor gets enormous amounts of data to train its system and they receive a ton of feedback on how well it functions can this be accomplished.

And unless you can conclusively demonstrate it in your business, avoid purchasing a system that is a black box. We must take the time to test these technologies in the real world and fine-tune them for best use, because management and hiring decisions in every firm are frequently based on culture.

For instance, IBM has invested years in tailoring its AI-based career and pay solutions to the needs of its business model. Now that they are supplying these tools to business clients, they have discovered that each application teaches IBM new things about the algorithms that improve them for that particular sector, culture, or organizational requirement.

Despite these difficulties and dangers, there is a tremendous advantage. Payroll accounts for 40–60% of a company’s revenue, and management decisions based on intuition account for a large portion of this massive expense.

We think productivity, performance, and employee health will significantly increase as AI technologies in HR get smarter, more reliable, and more

laser-focused on certain issues. All we need to do is wait, watch, and be prepared to put money into it.

3.4 IMPLICATIONS

It is no longer necessary to spend hundreds of man-hours sorting through tens of thousands of CVs and online job board profiles in search of new hires. In theory, a company that offers AI recruiting services can claim on their blog that they can “reduce time to hire from 34 days to 9 days” with the help of their AI candidate sourcing algorithm.

Utilizing a non-biased process that eliminates preconceptions from the sourcing process and identifies individuals who are technically qualified for the role, this represents a 73.53% increase in candidate sourcing and onboarding efficiency.

Another task that AI technology can automate is candidate screening. This sophisticated strategy allows for the collection of much more online data, including information from social media accounts, past online employment histories, and educational credentials, which improves the ranking system of prospects that recruiters can choose from (Hajimahmud et al., 2022).

The acceptance criteria established by the company and the candidate profile must meet in order to find and match the most promising individuals for the job, given that AI technology can effectively source and screen several candidates in a short amount of time (Khang et al., 2022a).

In order to discover the best applicant, AI systems can rate candidates according to a range of factors, using information such as experience, employment history, skill sets, and wage expectations. Because it can find passive applicants – who are typically the most sought after because they aren’t actively looking for new jobs and are content in their current role, meaning they are a benefit to their firm – this form of data processing is rising in value in the current market.

After you’ve found the perfect fit for the position, conducting interviews plays a significant role in the recruiting process. Many expatriates are actively looking for jobs abroad in today’s global labor market, but they are frequently unable to go to the country of their choice for a job interview.

A few businesses, including HireVue and Mya, are focused on developing AI interviewing software. By using video as the medium, many businesses are refocusing their efforts on streamlining the interviewing procedure. Premade questionnaires are used in applications like HireVue, and candidates can film themselves responding to them. This enables the recruiter to conduct general interviews with a greater number of possible applicants using prerecorded films, which can then be sorted through to choose people who will progress in the talent acquisition process.

The time required for hiring is significantly shortened because of AI technology, which handles all the legwork from candidate sourcing to interviews. As a result, the talent acquisition team is able to interact with potential employees to assess their suitability for the position in question and, inevitably, make a placement more quickly.

Many employees in larger companies may feel disengaged and lost by their employers and the human resources division. The tipping point for the majority of employees that leads to significant turnover rates might be anything from vacation

requests not being addressed to unenforced policies. An employee is not always content with a high income. Most individuals believe that their professions are the center of their lives, and who wants to have a miserable life? No one.

With more data at your disposal, you may improve the entire working environment for your staff by implementing the information and making the necessary changes to internal procedures or workplace culture. In a current tweet, Hughes states, “Excellent data is similar to good health. You take something for granted once you have it. You become anxious if you don’t.” Simply put, more data means more data analytics, which means more sophisticated insights that result in better working conditions for workers and lower turnover rates.

3.5 CONCLUSION

As much as AI continues to impact the HR technology environment, the developments in cognitive technology must be balanced with transparency by HR departments. To prevent unintentionally introducing bias into their programs, HR leaders and practitioners need to have a comprehensive understanding of how decisions are made. Making sure that employees trust the new technology will depend on this openness (Khang et al., 2023a).

AI should be utilized as a tool to advance our knowledge rather than as a problem solver or function to make difficult decisions for all HR functions. It will take some time for problems to be solved and advantages to be realized, just like with any new technology. Also, human intervention is required to assess AI decisions and special features. When deploying AI, it is crucial to focus on employee needs and potential results.

As you’ve probably already realized, there are many benefits to using AI technology for human resources and recruiting requirements (Misra et al., 2023).

However, nothing in life is flawless, particularly not a developing field of technology that hasn’t yet realized all of its promise. The availability of many services and programs that one might be used to in the manual, non-AI world, such as the ability to provide feedback to a candidate following a video interview or specific CV search criteria, may be lacking.

The ability to hand-pick candidates and get to know them personally throughout the recruitment process, which is currently impossible to accomplish while depending solely on AI technology, is one important feature that AI technology lacks. Humans perceive emotion, whereas AI perceives statistics, and this distinction won’t likely change in the near future.

As a result, it is up to you to evaluate whether AI technology is a good fit for your company’s needs. Do you work for a company with thousands of employees, spend a lot of money on hiring each year, and wish there was a way to expedite the procedure and save money? Are you a mom-and-pop business with fewer than 15 employees that takes pride in getting to know each worker as if they were a member of the family? The beauty of the artificial intelligence (AI) market is that there is a solution for any business, no matter how big or little, with various goals, budgets, and desires.

It is up to you to choose what is best for your company, whether that means adjusting to the rapidly expanding AI market, keeping things as they are, or gradually

making incremental adjustments. In either case, AI technology is only going to advance, and eventually AI will become the standard, making the conventional methods of hiring and managing human resources seem archaic (Morris et al., 2023).

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4 AI-Based Competency Model and Design in the Workforce Development System

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4.1 INTRODUCTION

Employees are now expected to be more competent, which has sparked a so-called “war for talents.” Even though the topic of competence management itself and the identification of future competence demands have recently drawn increased attention, there is still no way to deal with the change in competency requirements brought on by the Fourth Industrial Revolution (Khang et al., 2023).

According to Karwehl and Kauffeld (2021), there is a wide range of opinions and forecasts about the present digital transition, indicating that it is both very relevant to society and fraught with ambiguity. As a result, the need for capacity shifts rapidly in the economy, therefore there is no longer a need for an observant and planner rather than a capacity strategy, considering strategic planning and preparation and methods to develop competence when facing difficulties.

When competency frameworks are developed with such a high level of scientific and professional understanding, human capital managers may utilize them with confidence and comfort (Midhat Ali et al., 2021). Such a framework's benefit would be its general applicability to various forms of capital with either no or little alterations. The underlying methodology offers direction for changing the process as may be necessary to satisfy particular capital or organizational demands.

One may argue that competence frameworks should be tailored to each organization's culture, as well as its current and future demands (Semenova, Zelenyuk & Savinov, 2021). Generic competence frameworks are accessible, since they are necessary for such circumstances. As a result, the requirement for enterprises to address the competency portion is compelling and vital for human capital management (HCM). The majority of business owners are aware that of all the resources available to them, human resources are the most dynamic and demanding.

The business creates a human resource management (HRM) organization made up of corporate managers, and the supervisor and groups of stakeholders in charge of managing human resources put the idea dynamism into practice in a systematic and all-encompassing way (Sedyastuti et al., 2021). An efficient tool is required to aid managers in administration as the demand for human resource managers and the level of specialization of human resource departments both keep rising.

The importance of HCM for the growth of cultural procedures in modern society is quickly expanding because it is associated with the potential for novel strategic solutions to the serious challenges affecting civilization, such as resource scarcity and inequity, talent wars, the advancement of replacement and green technologies, climate change, etc. (Widarni & Wilantari, 2021).

Digital breakthroughs require specific professional capabilities from people with a strong position in the labor market who are in great demand (Choudhury, 2021). It is without a doubt an important contribution to the theory of human capital and the study of its role in the economy that competency and training are important factors in the growth and evolution of HCM (Dobrowolski, Drozdowski & Ledzianowski, 2021) as shown in [Figure 4.1](#).

Competence comprises knowledge, power, and aptitude. According to its origins, competence refers to a person's exceptional qualities and behavioral abilities as well as their knowledge, demeanor, and talents.

A capable leader is required for the growth of management to enhance HCM performance (Drozdowski, Rogozińska-Mitrut & Stasiak, 2021). A capable leader is required for the growth of management to enhance HCM performance (Drozdowski, Rogozińska-Mitrut & Stasiak, 2021). They presented a significant discussion on managerial competence in small businesses by assessing managerial strategies in order to set strategic edge and given an additional explanation about the significant relation between managerial competence and HCM performance. It will provide a



FIGURE 4.1 General framework of HCM.

clear recommendation as to whether management skills will play a different role or can indirectly affect HCM's performance (Snehal, Babasaheb & Khang, 2023).

HCM will prioritize investing in general competence training over particular competency training because the latter has no additional value outside of present employment and can raise a person's selling value in the labor market.

Because of this, every employee will take advantage of the possibilities he or she has to expand his general skills (Islam & Amin, 2021). Organizations now need to establish ecological policies and programs due to growing environmental awareness and the acceptance of international environmental protection standards. HRM provides ecologically responsible activities, including internet recruitment, safeguarding digital records, conducting extensive interviews and supporting documents, receiving feedback and coaching, and receiving incentives based on initiatives.

The primary goal is to determine if HRM can increase sustainable development by implementing the HRM concept in organizations (Mehta & Ali, 2021). They may even be ready to forego using their resources if they believe that doing so will increase the human capital's worth. Employers will choose to invest in the growth of particular abilities that solely benefit training providers and may even raise the perceived cost of an employee moving employment.

Employee competency has grown, now encompassing knowledge, ability, and aptitude (Dencik & Stevens, 2021). In the past, competence has been described as a person's exceptional qualities, knowledge, and behavioral abilities as well as their

demeanor and aptitude. An HCM's organizational competency is an essential tool for personal, professional, and organizational growth. Adapting to or influencing others in line with situational demands promotes social cohesiveness for improved living standards.

Competency knowledge in the business is being attentive to and aware of others' attitudes, which necessitates the capacity to instinctively enhance performance in light of this knowledge (Ozden & Guleryuz, 2022). Promoting competence is crucial, as the world of work requires a person's cognitive, emotional, and physical resources. It has been seen and quantified that people with high competence intelligence are more productive.

The assessment of human capital's competency, which serves as the foundation for adapting quickly to shifting operational situations, is based on the current dynamic between workers and external factors. Employee conduct and the environment have dynamic, bidirectional interaction. Human capital adjusts its operations to its surroundings (Tailor, Pareek & Khang, 2022).

Employees simultaneously affect several elements of the environment through their competency potential (Mkrttchian & Chernyshenko, 2021). In light of this, it is critical to regularly refresh employee competence capital to meet the expectations enforced by a dynamic environment. Each worker should cultivate their sensitivity to a high level of efficiency and use it, as a critical element of their brand is active, focused, and an effort to ensure organizational growth.

The key areas of competencies that endanger business resource management, as demonstrated by the polled managers, are to be identified by HCM. The research then aims to ascertain how participants' historical factors influence manufacturing resource management's competency in risk assessment.

Competency is a complex notion, making it challenging to define and categorize. Multidimensional constructions make up the management competencies. The multidimensionality of the study that was done relates to enterprise human capital management's risk competencies (Babasaheb, Sphurti & Khang, 2023).

Competence is frequently linked with other conceivable personal qualities, such as intellect, the capacity for learning, or dexterity, with real competence being shown in the completion of HCM job tasks. The three competency typologies—cognitive, functional, and behavioral—appeared in several H (Class H Certificate means any of the Certificates that collectively constitute the Class bearing the alphabetic class designation “H”) certification frameworks and showed a thorough and unifying methodology. Cognitive ability provides competence knowledge (Rana et al., 2021).

The level of comprehension of pertinent theoretical and procedural information better contextualized in work processes is what is being discussed (Douglas, 2021). “How and when to know” are answered by functional competency. It speaks to the degree to which one practices and acquires necessary work skills.

The solution to whom to know is behavioral competence. One of the most important resources in each organization with unique strategic goals to achieve is managerial competency. Finding out how managerial cognition, human capital management, and social capital serve as the foundations of better managerial capacities is the goal of managerial competency.

While competence is the organization's capacity to produce new technical competencies, formed by the combination of newly developed HCM resources, marketing competence serves recently developed markets (Khang et al., 2023).

Additionally, technical competence refers to an organization's power to create things of value with specified qualities in HCM, and consumer competence refers to an organization's capacity to service certain clients (Mwaro, Ogada & Cheruiyot, 2021).

Hence, we proposed a discriminate regressive artificial neural network (DR-ANN) for the effective competency model for hiring a skilled professional for the HCM system.

The remaining sections in the chapter are structured as follows. The problem statement is presented in [Section 4.3](#). The explanations of the proposed work are provided in [Section 4.4](#). [Section 4.5](#) has discussed the results. The proposed chapter's conclusion is presented in [Section 4.6](#).

4.2 PROBLEM STATEMENT

In today's society, HCM is becoming increasingly crucial to the growth of socioeconomic processes. Innovative and digital technologies need specialized professional skills from highly desired individuals who play a crucial role in the growth of HCM. Hiring talented employees is essential for the advancement of HCM.

This study suggests using a DR-ANN to hire qualified personnel and develop an efficient competence model for an HCM system (Khang, Rani & Sivaraman, 2022).

4.3 PROPOSED WORK

The development of HCM significantly relies on qualified professionals. This chapter seeks to enhance the development of HCM utilizing the proposed DR-ANN employing AI (Hajimahmud et al., 2022).

The HCM database serves as the foundation for gathering the dataset. The HCM data are preprocessed using normalization. In this part, a detailed explanation is provided in [Figure 4.2](#).

4.3.1 COLLECTION OF DATA

Numerous information software programs have been adopted for use in all stages of the so-called "hiring funnel," including procurement, scanning, assessing, and applicant selection. In addition to resumes, forums like LinkedIn and the expansion of social media sites devoted to certain professions, crafts, or sectors have played a key role in the creation of data sources that assist advertising, searches, and applicant filtering (Khang, Chowdhury & Sharma, 2022).

Predictive tools that rely on AI to carry out tasks such as categorization, scoring, ranking, or recommendations are frequently developed using this data (Bhambri et al., 2022).

4.3.2 DATA PREPROCESSING

Preprocessing transforms raw data into a format that computers can interpret and analyze as part of the data mining and data analysis process. The organization of

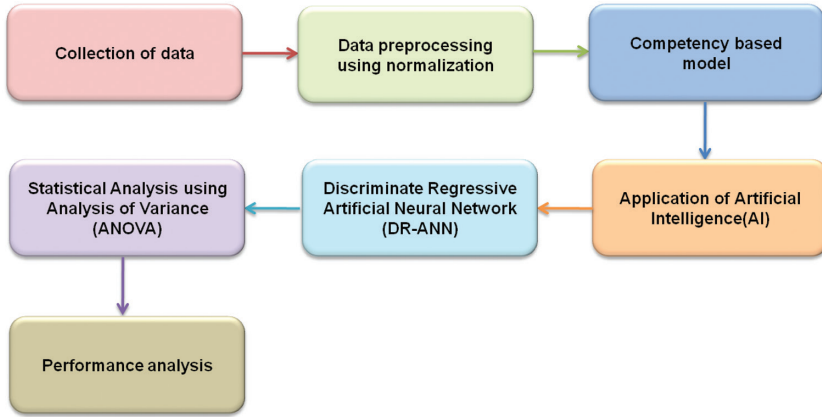


FIGURE 4.2 Flowchart of the proposed method.

the text, pictures, videos, and certain other authentic material is poor (Rani et al., 2023). It frequently has flaws and paradoxes, is difficult to understand, and lacks a cohesive design. A preprocessing step, a scaling approach, or a mapping technique is normalization.

The process of normalization converts unstructured data into normalized values or ranges of data using concepts like mean and standard deviation. As a result, the data information may be standardized using the normalization variable, as shown below:

$$u'_i = \frac{u_i - A}{\text{std}(A)} \quad (4.1)$$

$$\text{std}(A) = \sqrt{\frac{1}{(m-1)} \sum_{i=1}^m (u_i - \bar{A})^2} \quad (4.2)$$

$$\bar{A} = \frac{1}{m} \sum_{i=1}^m u_i \quad (4.3)$$

As a result, the normalization technique may be used to calculate the normalized values in each of the rows above. If all of the values in a row are the same and all of the variables are set to 0, the standard deviation of that row is 0. The normalization method indicates the range of values between 0 and 1. A technique called scaling offers a range of values between -1 and 1.

4.3.3 COMPETENCY-BASED MODEL

A competence model is a group of skills that collectively determine effective performance in a certain work environment. The cornerstones of crucial human

capital activities, including recruiting, firing, learning and support, and performance appraisal, are competency models.

When critical work roles, positions, or other job designations are identified by the organization's executives as needing recruiting activities, recruitment can commence the competency model. Setting priorities is required for this.

The length of time the recruiting process will last must also be decided by the decision-makers. A competency-based approach to hiring and choosing the roles or positions that are most essential to the organization's performance should be the focus of the organization's available resources.

Leaders can assess the costs and advantages of extending after important jobs have been filled. The leaders of the company must verify the correctness of the job descriptions and requirements once they have determined their recruiting needs. The data should satisfy the following criteria and can be obtained by modest to intensive job studies, as shown in [Figure 4.3](#).

The key benefits of competency-based model for the recruitment of qualified employees for HCM are as follows:

- The managers who are looking for candidates agree on the work outputs, activities, and tasks, as well as the job skills and the behavioral indicators by which those competencies may be measured.
- Competency-based systems eliminate bias, yet despite this, it's been stated that one of the main barriers preventing managers from embracing diversity recruiting programs is their concern that bringing in individuals who may not share their viewpoints will reduce productivity.



FIGURE 4.3 Competency-based recruitment.

- The managers looking for candidates have defined and evaluated the key qualities that are the best indicators of work success.
- Systems based on competencies forecast upcoming achievements by analyzing previous or present professionals who were effective in that capacity, and core skills are determined for certain roles. Critical occurrences, choices, and activities are assessed through behavior management to identify essential capabilities, which are then included in a predictive model to guide the talent acquisition process.
- For the aforementioned goals, the competencies required for effective job performance are still relevant.

4.3.4 APPLICATIONS OF ARTIFICIAL INTELLIGENCE

AI can modify the position of the hiring manager through augmented intelligence, allowing managers to be more responsive in their recruitment, assist in determining whether an applicant fits into the corporate culture, and improve their connections with hiring managers by using data to measure things like quality of hire (Khang et al., 2023).

AI-powered software is capable of analyzing users' internet accounts on social media and other publicly available information. Based on this information, the technology can forecast the future (Rani et al., 2022). For instance, what kind of positions could someone be interested in, and how likely are they to take a job? It can even examine the resumes of applicants who have already begun working for the organization. The program can locate people with comparable attitudes and skill sets by merging all the information at its command. It can determine which people could be interested in particular positions and target them specifically with appropriate job advertisements.

Managers don't want the judgment to be tainted by bias while trying to identify the ideal applicant for the position. Furthermore, there are several intriguing AI uses in hiring that might lessen prejudice. There are resources available to assist recruiters in creating objective job advertisements, as shown in [Figure 4.4](#).

Most recruiters are quite busy. The more time-consuming and difficult procedures can be automated in this regard, so AI is more beneficial in employee recruitment (Khanh & Khang, 2021).

4.3.5 DISCRIMINATE REGRESSIVE ARTIFICIAL NEURAL NETWORK

The DR-ANN model was based on a strategy that had been employed with great success in gene profiling using employee data in the past. DR-ANN approaches have subsequently been used in a broad variety of sectors and have proven successful at handling challenging issues. The strategies have been used in a corporate environment to hire employees, oversee advancement, forecast credit grades, and track a company's stock performances (Rani et al., 2021).

DR-ANN approaches outperformed popular multidimensional quantitative tools in several situations. The employee data were divided into three groups by chance before artificial neural network (ANN) training: 60% for training, 20% for

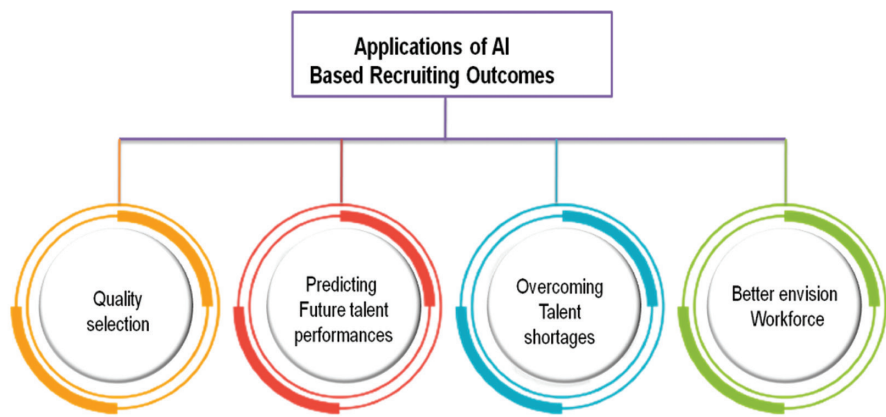


FIGURE 4.4 AI-based recruitment.

validation, and 20% for independently testing the model on blind data (Bhambri et al., 2022).

The process employed discriminate regression, which has been demonstrated to perform better and be more reliable than other approaches like leave-one-out cross-validation. Additionally, it prevents the employee data from being too fit, a back-propagation technique and a multilayer perception design were used in the modeling of artificial neural networks (Rani et al., 2023).

The sigmoid transfer function utilized in this technique allowed for the updating of weights based on feedback from mistakes. The movements in which a percentage of the prior shift in rates was pushed back to the present change in weights were set at 1, while the learning rate at which weights are updated as a fraction of the error was set at 0.5.

A single hidden layer of the artificial neural network design had two hidden nodes. The highest number of epochs utilized was 400, and the testing allowed for a range of 70 epochs without growth. This served to avoid the model from overfitting.

Results for the typical test performance and typical test error were given. The percentage of test cases that are successfully anticipated is shown by the average test performance. The root means square error for the test employee data set, which represents the discrepancy between the model’s estimated parameters and the testing data collection’s measured data, is the overall average error.

A technique of discrimination regression incorporates ANN to adjust for all pertinent job seeker traits that are available to recruiters and surveillance of recruiters’ online job search behavior.

According to this technique, which uses the public employment service’s online recruiting platform, people from minority and immigrant ethnic groups are more likely to be contacted by recruiters than people from the majority group. The

hiring managers have less time to evaluate their candidates' qualifications from various groups.

Recruiters and policymakers may use the methodology to continuously hire without discrimination, identify some of the sources of prejudice, and guide steps to tackle it. It is generally applicable, nonintrusive, and cost-effective.

4.3.6 STATISTICAL ANALYSIS BY ANALYSIS OF VARIANCE

Analysis of variance (ANOVA) is a statistical method used to determine if the results of three or more organizations differ from one another substantially. ANOVA compares the means of tested concentrations to examine the influence of one or more factors.

ANOVA statistics show that the suggested DR-ANN procedure enables the HCM to choose the best employees following the needs and specifics of each profession as determined through the job role and work specifications.

DR-ANN can forecast employee performance, as well as how to compensate each employee based on performance, and emphasizes that one of the key elements influencing employee performance and productivity is the talent acquisition process.

The information, skills, talents, and experience that are essential for successful and high-level recruiting are identified by DR-ANN. It may design standards for performance assessment, plan and create objectives for employee training and HCM development programs, and provide criteria for selection, hiring, and development of HCM.

4.3.7 IMPROVED SELF-ADAPTIVE FROG LEAPING ALGORITHM

The improved self-adaptive frog leaping algorithm (ISAFILA) is an artificial intelligence-based periodic basic search approach that was developed by simulating the information sharing and communication mechanisms in frog searching. It offers fewer parameter settings, quick solution speed, and great global optimization capability. There are mainly three types of operations involved in ISAFILA. They are

- Initialization operation
- Grouping operation
- Local update operation

4.4 PERFORMANCE ANALYTICS

Employee competence has increased, and it now includes knowledge, power, and aptitude. Historically, competence has been defined as an employee's remarkable traits, behavioral skills, and knowledge, as well as their attitude, manner, and ability. In this chapter, the competency-based hiring process is designed using DR-ANN with AI for the recruitment of qualified employees in crucial situations of HCM.

The effectiveness of the suggested DR-ANN strategy in assuring the selection process is examined in this section. The proposed system can give accuracy, hiring time, selection of skilled employees, and manager feedback for HCM. The existing methods are the Bayesian tuned Gaussian process regression (BT-GPR), avatar-based control (A-BC), organizational resilience (OR), and ensemble neural network (ENN).

4.4.1 ACCURACY

When using competence models to evaluate job candidates, accuracy is one criterion to take into account. The phrase “degree to which the outcome of a measurement agrees with the appropriate value or a standard” is used to define accuracy.

The percentage of forecasts that the model correctly predicted is known as accuracy. Every enterprise needing data classification has to do an accuracy evaluation. The method of choosing the approach is contrasted with that of other techniques.

Figure 4.5 shows the existing model and the recommended model accuracy. The recommended process is quite effective in choosing applicants. The BT-GPR achieved 55%, the A-BC achieved 64%, the OR achieved 85%, the ENN achieved 74%, and the recommended method achieved 95%.

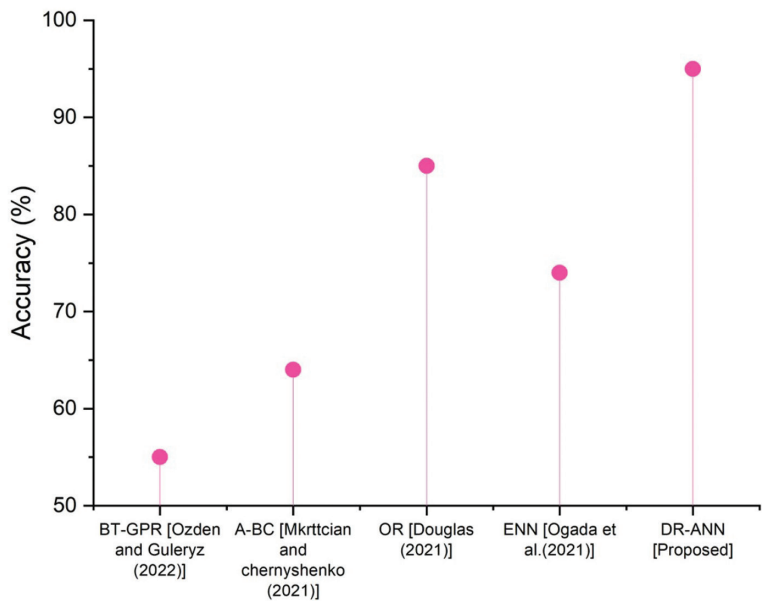


FIGURE 4.5 Accuracy of the existing and proposed model.

4.4.2 HIRING TIME

The term “hiring time” describes the amount of time it takes an enterprise or recruiting team to identify a qualified applicant and fill a vacancy in an organization. The “hiring time” meter gauges how quickly an applicant is evaluated, evaluated, and hired for a position.

A lengthy hiring process suggests a sluggish, ineffective procedure with needless obstacles. It immediately affects the success of the HCM and the capacity to recruit the most talented individuals, making it one of the most important recruiting indicators.

Figure 4.6 shows the hiring times for the proposed and existing models. DR-ANN takes less time to hire than existing models. The BT-GPR achieved 90 days, the A-BC achieved 65 days, the OR achieved 74 days, the ENN achieved 85 days, and the recommended method achieved 50 days.

4.4.3 SELECTION OF SKILLED EMPLOYEE

The recruiting and selection process involves a number of processes. They consist of posting job openings and sourcing applicants, looking through applications, selecting candidates, interviewing candidates, and making an offer. Throughout the interviewing process, HR collaborates closely with recruiting managers.

The only people who truly understand what a role entails and the sorts of talents a person needs to perform the job successfully are managers. Employers should pay

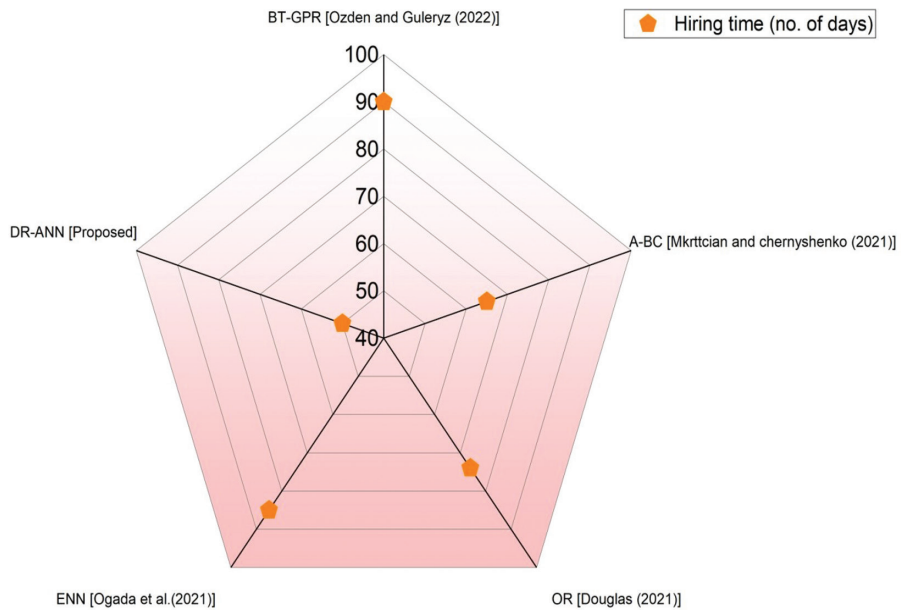


FIGURE 4.6 The hiring time of the existing and proposed model.

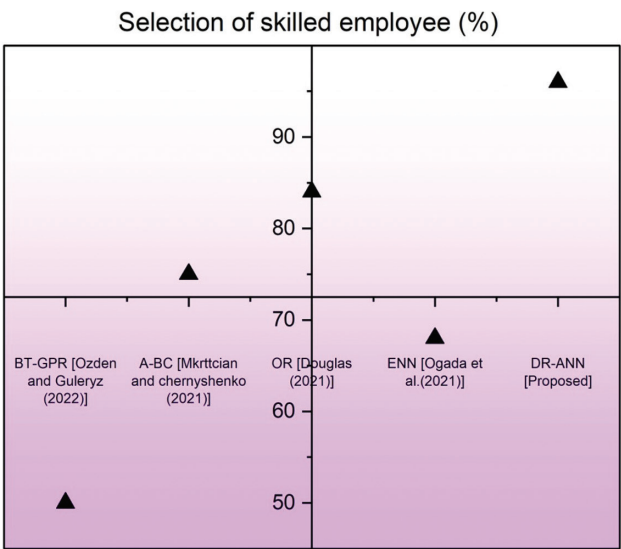


FIGURE 4.7 Selection of skilled employees in the existing and proposed model.

close attention to the selection of skilled employees, since it can increase retention and reduce turnover. Employers want to hire the best person for the job when they are looking to fill a position.

Figure 4.7 depicts the selection procedure for the existing and suggested models. DR-ANN provides a more thorough selection of qualified candidates than existing approaches. The BT-GPR scored 50%, the A-BC scored 75%, the OR scored 84%, the ENN scored 68%, and the recommended method scored 96%.

4.4.4 MANAGING THE FEEDBACK

One aspect to take into account while assessing the recruiting process through the AI-based competency model in HCM is manager feedback. Feedback is useful to everyone. We develop and change by stepping outside of the social circles with critical feedback. The selecting procedure is comparable in this regard.

Without asking for input from managers, the recruitment team won't ever know where it can be enhanced. Because it provides us with a neutral standpoint on what it's like to apply for a job with their organization, gathering input based on the selection process is crucial. It would be crazy not to take advantage of the fact that these individuals have experience on the other side of the hiring process. The feedback from managers indicates how closely the outcome of a measurement matches the appropriate personnel hire or selection.

Figure 4.8 shows a comparison of the suggested approach's selection with that of other methods. The recommended process for choosing employees has received

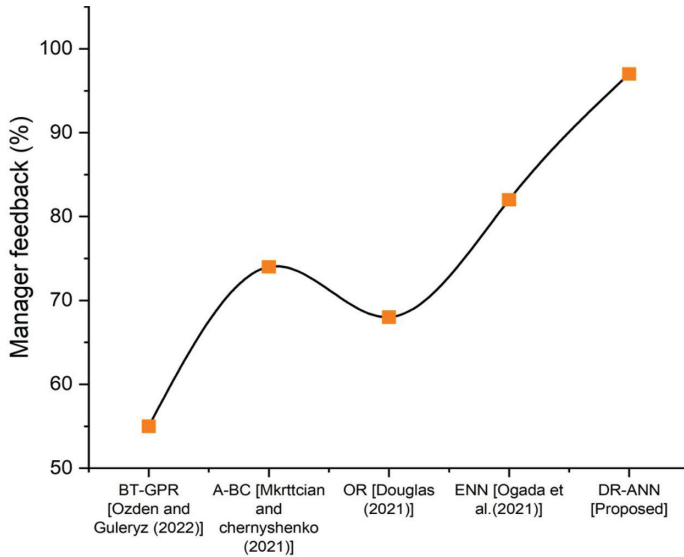


FIGURE 4.8 Manager feedback on an existing and proposed model.

extremely positive feedback from the managers. The BT-GPR received 55%, the A-BC received 74%, the OR received 68%, the ENN received 82%, and the recommended method received 97%.

4.4.5 DESIGN PRINCIPLE OF COMPETENCY MODEL

4.4.5.1 Excellent Communication

Outstanding communication being able to communicate ideas clearly and precisely in a form or language that clients can grasp actively engages in communication, it demonstrates patience, and is upbeat in all circumstances when answers all inquiries completely while providing the client with the best possible service.

4.4.5.2 Effective Teamwork

The employee possesses the capacity to work both independently and collaboratively while maintaining a balance, and is always willing to help coworkers if they need it. The employee respects the contributions of all team members, provides the team with the knowledge and resources they need, and has integrity and high ethical standards.

The employee has moral character both inside and outside the workplace by acting in a way that upholds the reputation of the business in every circumstance.

4.4.5.3 Culture Sense

The employee respects all clients and coworkers regardless of their cultural differences (e.g., ethnicity, religion, sex, nationality), adapts to many cultural contexts,

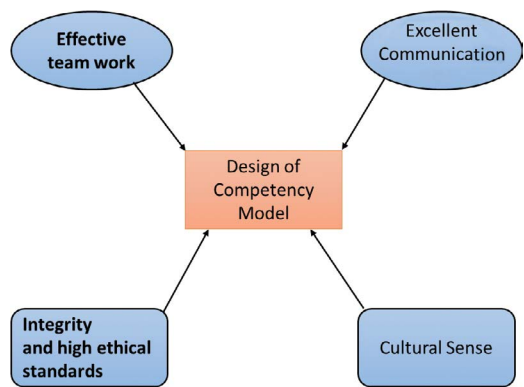


FIGURE 4.9 Design principles of the competency model.

and handles them well. The employee offers solutions and preventative measures for potential issues caused by cultural differences as [Figure 4.9](#).

4.5 CONCLUSION

Regardless of the industry and company operations, employee selection is a crucial activity in the field of HCM and is essential to the success of every enterprise. However, this becomes much more crucial when choosing the appropriate candidates for professional positions, such as employees. As a result, a more methodical approach to hiring skilled workers that are based on high criteria of ability is required.

For professional selection in HCM, this chapter offered a novel AI-based DR-ANN. Utilizing ANOVA, the competency model is analyzed. The results show that the recommended DR-ANN technique is effective in terms of accuracy, hiring time, selection of skilled employees, and fees. Manager feedback back of managers and that it beats the classic BT-GPR, A-BC, OR, and ENN algorithm methods.

Future research on the suggested study may focus on improving HCM operational behavior and the employment of varied applicants (Khang, Rani & Shah, 2024).

In the future, we may look into introducing optimization approaches to boost additional performance measures and enhancing the recruiting of qualified employees in these areas using a variety of techniques (Khang et al., 2023).

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5 Data

An Anchor for Decision-Making to Build the Future Workforce Management System

Samiksha Shukla, Aerica Rishiraj, and Yash Lucas

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